SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: CSA1402 **COURSE NAME: Compiler Design for High-Level Programming**

COURSE OUTCOME COVERED

CO3: *Analyze syntax-directed translation method using synthesized and inherited attributes to generate a Parser Tree. (BL4)*

Assessment Tool Weightage: 10%

QUESTIONS: 60 **Total Marks:60**

Annexure A

MCQ

Code of Conduct:

I A. Midhuna/192521302 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Midhuna. A

Criteria	Weightage	Level 4 – Exemplary (50–60 Marks)	Level 3 – Proficient (40–49 Marks)	Level 2 – Developing (30–39 Marks)	Level 1 – Beginning (0–29 Marks)
Number of Correct Answers	100%	50 to 60 correct answers	40 to 49 correct answers	30 to 39 correct answers	0 to 29 correct answers
Accuracy	100%	Excellent (90–100%)	Good (75–89%)	Average (50–74%)	Poor (Below 50%)

Course Faculty

Annexure A

MCQ

1. A compiler designer wants to attach semantic meaning to grammar productions in order to generate intermediate representations efficiently. Which compiler concept is most suitable for this purpose?

A) Lexical buffering
B) Associating semantic rules with grammar productions
C) Register allocation
D) Peephole optimization

ANS: B) Associating semantic rules with grammar productions

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2. While compiling an arithmetic expression, a compiler internally creates a hierarchical representation showing operators and operands. Which structure is being generated?

A) Stack frame
B) Queue structure
C) Syntax Tree
D) Linked List

ANS: C) Syntax Tree

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3. A compiler evaluates attributes from child nodes before computing the parent node value during semantic analysis. Which type of attribute evaluation is taking place?

A) Parent-driven evaluation
B) Synthesized attribute evaluation
C) Inherited attribute evaluation
D) Dynamic attribute evaluation

ANS: B) Synthesized attribute evaluation

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4. A language processor uses LR parsing to evaluate semantic information during reductions. Which attribute grammar is naturally suitable in this case?

A) Context-free attribute grammar
B) Recursive attribute grammar
C) S-attributed definition
D) Circular attribute grammar

ANS: C) S-attributed definition

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5. During semantic rule evaluation, a parser requires both parent information and child information while traversing grammar symbols from left to right. Which definition supports this requirement?

A) S-attributed definition
B) Static semantic definition
C) L-attributed definition
D) Ambiguous definition

ANS: C) L-attributed definition

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6. A compiler engineer wants child nodes to be processed before the parent node during semantic evaluation of expressions. Which traversal strategy should be selected?

A) Preorder traversal
B) Inorder traversal
C) Postorder traversal
D) Level-order traversal

ANS: C) Postorder traversal

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7. An expression such as `(a+b)*c` is represented internally without unnecessary grammar details. What does this representation mainly capture?

A) Memory layout

B) Hierarchical program structure

C) Register allocation

D) CPU execution order

ANS: B) Hierarchical program structure

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8. While evaluating semantic rules, information is passed from a parent node toward its child nodes. Which type of attribute is involved?

A) Synthesized attribute

B) Terminal attribute

C) Inherited attribute

D) Dynamic attribute

ANS: C) Inherited attribute

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9. A compiler rejects the statement `int x = "hello";` before execution begins. Which feature of the programming language is responsible for detecting this issue?

A) Token generation

B) Type system

C) Syntax optimization

D) Loop unrolling

ANS: B) Type system

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10. Consider the statement `float value = 25;`. Which compiler activity occurs automatically here?

A) Explicit casting

B) Dynamic conversion

C) Implicit type conversion

D) Symbol replacement

ANS: C) Implicit type conversion

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11. A semantic analyzer verifies whether variables are declared and operations are type-compatible. Which compiler phase performs these tasks?

A) Lexical Analysis
B) Syntax Analysis
C) Semantic Analysis
D) Linking

ANS: C) Semantic Analysis

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12. A compiler developer chooses a representation that eliminates unnecessary grammar symbols while preserving expression meaning. Which structure is preferred over a parse tree?

A) Symbol Table
B) Syntax Tree
C) Token Stream
D) Finite Automaton

ANS: B) Syntax Tree

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13. During compilation, all information regarding identifiers, scope, and datatype is maintained centrally. Which compiler component performs this responsibility?

A) Code Generator
B) Symbol Table
C) Syntax Parser
D) Optimizer

ANS: B) Symbol Table

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14. A compiler uses only synthesized information for semantic evaluation of grammar productions. Which type of attribute grammar is being implemented?

A) Dynamic grammar

B) L-attributed grammar

C) S-attributed grammar

D) Context-sensitive grammar

ANS: C) S-attributed grammar

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15. A recursive descent parser performs semantic translation while expanding productions from the start symbol downward. Which translation approach is being followed?

A) Bottom-up translation

B) Top-down translation

C) Shift-reduce translation

D) Operator precedence translation

ANS: B) Top-down translation

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16. During shift-reduce parsing, intermediate grammar symbols are temporarily stored before reductions occur. Which data structure supports this operation most effectively?

A) Queue

B) Heap

C) Stack

D) Linked List

ANS: C) Stack

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17. A compiler reports an error when a boolean value is added to an integer variable. What compiler mechanism detected this problem?

A) Syntax checking

B) Type checking

C) Token validation

D) Memory management

ANS: B) Type checking

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18. A programmer declares a variable to store whole numbers without decimal values. Which datatype is most appropriate?

A) Boolean

B) Integer

C) Character

D) String

ANS: B) Integer

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19. Two array types are considered equivalent because they possess identical internal structures even though their names differ. Which type equivalence principle is applied?

A) Name equivalence

B) Structural equivalence

C) Functional equivalence

D) Dynamic equivalence

ANS: B) Structural equivalence

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20. A compiler implementing LR parsing evaluates semantic rules during reductions. Which evaluation strategy is best suited for this implementation?

A) Top-down evaluation

B) Parallel evaluation

C) Bottom-up evaluation

D) Recursive evaluation

ANS: C) Bottom-up evaluation

21. During expression analysis, a compiler stores operators as internal nodes and operands as leaves. Which representation is being used?

A) AVL Tree
B) Heap Tree
C) Syntax Tree
D) B-Tree

ANS: C) Syntax Tree

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22. A semantic rule requires information from symbols appearing to the left side of a production. Which attribute grammar supports such evaluation?

A) S-attributed grammar
B) L-attributed grammar
C) Ambiguous grammar
D) Recursive grammar

ANS: B) L-attributed grammar

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23. After semantic analysis, a compiler converts validated source constructs into machine-independent instructions. Which compiler phase performs this task?

A) Lexical Analysis
B) Intermediate Code Generation
C) Linking
D) Parsing

ANS: B) Intermediate Code Generation

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24. A programmer intentionally converts a floating-point value into an integer using a language construct. What is this process called?

A) Promotion

B) Reduction

C) Casting

D) Coercion

ANS: C) Casting

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25. While scanning source code, a compiler groups characters into meaningful units such as keywords and identifiers. Which phase performs this operation?

A) Semantic Analysis

B) Syntax Analysis

C) Lexical Analysis

D) Optimization

ANS: C) Lexical Analysis

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26. A parse tree generated for a program statement explicitly contains all grammar symbols used in derivation. What is the primary characteristic of this structure?

A) Contains machine instructions

B) Represents grammar symbols

C) Removes terminals

D) Stores registers only

ANS: B) Represents grammar symbols

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27. A bottom-up parser starts constructing derivations beginning with identifiers and constants before reaching the start symbol. Which nodes are processed first?

A) Root nodes

B) Internal nodes

C) Leaf nodes

D) Header nodes

ANS: C) Leaf nodes

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28. During assignment checking, a compiler ensures the datatype of the expression matches the datatype of the variable receiving it. Which component performs this verification?

A) Loader
B) Semantic checker
C) Tokenizer
D) Syntax formatter

ANS: B) Semantic checker

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29. Two variables are treated as equivalent only because they were declared using the same type name. Which equivalence mechanism is being followed?

A) Structural equivalence
B) Dynamic equivalence
C) Name equivalence
D) Functional equivalence

ANS: C) Name equivalence

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30. A semantic evaluator computes all child expressions before processing the parent expression. Which traversal technique reflects this behavior?

A) Preorder traversal
B) Inorder traversal
C) Postorder traversal
D) Reverse traversal

ANS: C) Postorder traversal

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31. A compiler designer inserts executable semantic instructions directly inside grammar productions to perform translation tasks. What is this mechanism called?

A) Syntax-directed translation scheme
B) Token generation mechanism
C) Recursive grammar scheme
D) DFA representation

ANS: A) Syntax-directed translation scheme

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32. A parser repeatedly performs shift and reduce operations until the start symbol is produced. Which parsing strategy is being used?

A) Predictive parsing
B) Top-down parsing
C) Bottom-up parsing
D) Recursive parsing

ANS: C) Bottom-up parsing

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33. A strongly typed programming language prevents invalid arithmetic operations between incompatible datatypes. Which language feature enforces this?

A) Type system
B) Code optimizer
C) Symbol allocator
D) Loader

ANS: A) Type system

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34. A programmer creates a collection capable of storing multiple values of the same datatype under one name. Which datatype category does this belong to?

A) Primitive datatype
B) Derived datatype

C) Boolean datatype
D) Character datatype

ANS: B) Derived datatype

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35. In a syntax tree representing an assignment statement, the highest node corresponds to the complete statement meaning. Which element does the root node generally represent?

A) Constant value
B) Identifier name
C) Entire expression or statement
D) Memory address

ANS: C) Entire expression or statement

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36. A compiler reports an error because a variable is used outside its declared scope. Which compiler phase detects this issue?

A) Syntax Analysis
B) Semantic Analysis
C) Code Optimization
D) Linking

ANS: B) Semantic Analysis

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37. A compiler automatically converts an integer operand into a floating-point operand before arithmetic evaluation. What type of conversion is this?

A) Narrowing conversion
B) Implicit widening conversion
C) Explicit casting
D) Recursive conversion

ANS: B) Implicit widening conversion

38. During attribute evaluation, semantic rules are processed from left to right so that inherited information becomes available when required. Which attribute grammar follows this restriction?

A) Circular grammar

B) L-attributed grammar

C) S-attributed grammar

D) Ambiguous grammar

ANS: B) L-attributed grammar

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39. A compiler uses shift-reduce actions and reduction tables to parse complex programming constructs. Which parser is most likely being used?

A) Predictive parser

B) Recursive descent parser

C) LR parser

D) LL (1) parser

ANS: C) LR parser

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40. A compiler engineer associates semantic computations directly with grammar rules for translation purposes. To what are these semantic rules attached?

A) Tokens only

B) Grammar productions

C) Machine registers

D) Memory blocks

ANS: B) Grammar productions

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41. During semantic evaluation, a node computes its value entirely using values obtained from child nodes. Which attribute type is involved?

A) Inherited attribute

B) Dynamic attribute

C) Synthesized attribute

D) External attribute

ANS: C) Synthesized attribute

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42. A compiler designer prefers a compact internal representation that removes unnecessary non-terminal symbols from derivations. Which representation is best suited?

A) Parse Tree

B) Syntax Tree

C) NFA

D) Transition Graph

ANS: B) Syntax Tree

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43. A program contains the statement `total = amount + tax;` where `tax` has never been declared. Which compiler phase identifies this error?

A) Lexical Analysis

B) Parsing

C) Semantic Analysis

D) Code Generation

ANS: C) Semantic Analysis

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44. A compiler automatically changes the datatype of an operand during expression evaluation to avoid type mismatch. What is this process called?

A) Coercion

B) Parsing

C) Reduction

D) Folding

ANS: A) Coercion

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45. Two structures are treated as equivalent because they contain identical field arrangements and datatypes. Which equivalence principle is being applied?

A) Name equivalence

B) Scope equivalence

C) Structural equivalence

D) Dynamic equivalence

ANS: C) Structural equivalence

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46. A compiler implementing LR parsing evaluates semantic actions during reduce operations. Which evaluation strategy aligns best with this parser behavior?

A) Predictive evaluation

B) Recursive evaluation

C) Bottom-up evaluation

D) Top-down evaluation

ANS: C) Bottom-up evaluation

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47. A compiler developer chooses an attribute grammar that can be evaluated naturally during reduce actions without inherited information. Which grammar type is most appropriate?

A) L-attributed grammar

B) S-attributed grammar

C) Ambiguous grammar

D) Circular grammar

ANS: B) S-attributed grammar

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48. In a syntax tree for arithmetic expressions, the terminal values such as variables and constants appear at which location?

A) Internal nodes
B) Root node
C) Leaf nodes
D) Header nodes

ANS: C) Leaf nodes

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49. A programmer explicitly converts a floating-point result into an integer before storing it in a variable. Which operation is being applied?

A) Promotion
B) Casting
C) Translation
D) Coercion

ANS: B) Casting

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50. After successful syntax checking, a compiler begins verifying meanings, datatypes, and scope rules. Which phase starts next?

A) Lexical Analysis
B) Semantic Analysis
C) Linking
D) Loading

ANS: B) Semantic Analysis

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51. A programming language verifies datatype compatibility during compilation and sometimes again during execution. What does this indicate about type checking?

A) It occurs only during loading

B) It occurs only during runtime

C) It can occur during compile time and runtime

D) It is unnecessary in modern languages

ANS: C) It can occur during compile time and runtime

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52. A parser constructs derivations beginning from tokens and gradually reduces them until the start symbol is obtained. Which parsing approach is demonstrated?

A) Top-down parsing

B) Predictive parsing

C) Bottom-up parsing

D) Recursive parsing

ANS: C) Bottom-up parsing

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53. A compiler internally represents expressions using parent-child relationships between operators and operands instead of linear text order. Which representation style is being followed?

A) Sequential representation

B) Hierarchical representation

C) Tabular representation

D) Circular representation

ANS: B) Hierarchical representation

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54. A compiler phase determines whether expressions, assignments, and declarations carry valid meaning according to language rules. Which aspect is primarily analyzed?

A) Syntax structure

B) Meaning of constructs

C) Register allocation

D) Instruction scheduling

ANS: B) Meaning of constructs

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55. Two datatype declarations are accepted as compatible because the language rules consider them logically interchangeable. What does this describe?

A) Operator precedence

B) Variable scope

C) Type equivalence

D) Memory allocation

ANS: C) Type equivalence

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (90–100 Marks)	Level 3 – Proficient (75–89 Marks)	Level 2 – Developing (50–74 Marks)	Level 1 – Beginning (0–49 Marks)
Number of Correct Answers	100%	90 to 100 correct answers	75 to 89 correct answers	50 to 74 correct answers	0 to 49 correct answers

Criteria	Weightage (%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Conceptual Understanding	30	Demonstrates clear and complete understanding of the concept	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor or incorrect understanding
Accuracy of Answer	25	Completely correct answer with no errors	Mostly correct with minor errors	Partially correct with noticeable errors	Incorrect or irrelevant answer
Application of Knowledge	20	Correctly applies concepts to solve the question/problem	Applies concepts with minor mistakes	Limited or partially correct application	No or incorrect application
Completeness of Response	15	Covers all required parts of the question	Covers most parts with minor omissions	Covers only some parts	Incomplete or missing response
Clarity & Presentation	10	Clear, concise, and well-	Understandable with	Somewhat unclear or	Difficult to understand

Criteria	Weight age (%)	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
		organized answer	minor issues	poorly structured	